

Balance-Law Finite-Size SPA Waveform Notes

May 4, 2026

1 Purpose

This note documents the finite-size waveform currently implemented in `src/finite_size/waveform.py`. The important update is that the stationary-phase approximation (SPA) is not constructed by differentiating a pre-written closed-form phase. Instead, the code starts from the binding energy and flux structure given in Appendix A of arXiv:2410.00294, solves the balance-law chirp rate, and then builds both the Fourier phase and Fourier amplitude from that same chirp rate.

The model is still an inspiral-only benchmark target. It is not the full `IMRPhenomD+FiniteSize` implementation used in the paper. The goal is a small, reproducible waveform that tests whether an agent can translate finite size theory content into a working forecast waveform.

2 Conventions

The code uses detector-frame masses. The chirp mass M_c is input in solar masses, the luminosity distance d_L is input in Mpc, and the total mass is

$$M = \frac{M_c}{\eta^{3/5}}. \quad (1)$$

In seconds,

$$M_{\text{sec}} = M M_{\odot}, \quad M_{\odot} = 4.925491025543576 \times 10^{-6} \text{ s}. \quad (2)$$

The PN velocity variable is

$$v = (\pi M_{\text{sec}} f)^{1/3}. \quad (3)$$

The symmetric and antisymmetric spin combinations are

$$\chi_s = \frac{\chi_1 + \chi_2}{2}, \quad \chi_a = \frac{\chi_1 - \chi_2}{2}, \quad \delta = \sqrt{1 - 4\eta}. \quad (4)$$

The final Fourier-domain waveform convention is

$$\tilde{h}(f) = A(f) \exp[-i\psi(f)], \quad \psi(f) = 2\pi f t_c - \phi_c - \frac{\pi}{4} + \psi_{\text{bal}}(f). \quad (5)$$

3 Energy Balance to SPA

Appendix A of arXiv:2410.00294 starts from the balance equation

$$-\mathcal{F}_{\infty}(v) - \dot{M}(v) = \frac{d\mathcal{E}}{dt}. \quad (6)$$

Writing the binding energy as $\mathcal{E} = M e(v)$, with M constant over the inspiral benchmark, Eq. (6) gives

$$\frac{dt}{dv} = -M_{\text{sec}} \frac{de/dv}{\mathcal{F}_{\infty}(v) + \dot{M}(v)}. \quad (7)$$

For the dominant $(\ell, m) = (2, 2)$ mode, $f = v^3/(\pi M_{\text{sec}})$. Hence

$$\dot{f} = \frac{df}{dt} = \frac{3v^2}{\pi M_{\text{sec}}} \left(\frac{dt}{dv} \right)^{-1}. \quad (8)$$

The gravitational-wave phase obeys

$$\frac{d\Phi_{\text{GW}}}{dv} = 2 \frac{v^3}{M_{\text{sec}}} \frac{dt}{dv}. \quad (9)$$

The code chooses a reference point v_{ref} at the taper cutoff and integrates backwards from that point:

$$T(v) = \int_v^{v_{\text{ref}}} \frac{dt}{dv'} dv', \quad (10)$$

$$\Delta\Phi_{\text{GW}}(v) = \int_v^{v_{\text{ref}}} 2 \frac{v'^3}{M_{\text{sec}}} \frac{dt}{dv'} dv'. \quad (11)$$

Up to arbitrary constant and linear-in-frequency pieces, which are absorbed by ϕ_c and t_c , the intrinsic SPA phase is

$$\psi_{\text{bal}}(v) = \Delta\Phi_{\text{GW}}(v) - 2\pi f(v) T(v). \quad (12)$$

In the Newtonian limit,

$$e(v) = -\frac{\eta v^2}{2}, \quad \mathcal{F}_{\infty}(v) = \frac{32}{5} \eta^2 v^{10}, \quad (13)$$

Eqs. (7)–(12) recover the usual curvature

$$\psi_{\text{bal}}(f) = \frac{3}{128\eta} v^{-5} + \text{constant} + \text{linear in } f. \quad (14)$$

4 SPA Amplitude

The time-domain amplitude is kept at restricted Newtonian order, but the SPA Jacobian is computed from the same chirp rate as the phase. The standard Newtonian TaylorF2 amplitude is

$$A_N(f) = \sqrt{\frac{5}{24}} \frac{M_{c,\text{sec}}^{5/6}}{d_{L,\text{sec}} \pi^{2/3}} f^{-7/6}. \quad (15)$$

The current balance-law amplitude is

$$A(f) = A_N(f) \left[\frac{\dot{f}_N}{\dot{f}} \right]^{1/2}, \quad \dot{f}_N = \frac{96}{5\pi M_{\text{sec}}^2} \eta v^{11}. \quad (16)$$

Thus the amplitude reduces exactly to A_N in the Newtonian limit and changes only through the finite-size correction to the chirp rate.

5 Binding Energy Used in the Code

The paper writes the binding energy as

$$\mathcal{E}(v) = -\frac{M\eta v^2}{2} \left[\mathcal{E}_{\text{NS}} + \mathcal{E}_{\text{SO}}v^3 + \mathcal{E}_{\text{SS}}v^4 + \mathcal{E}_{\text{SSS}}v^7 + \mathcal{E}_{\text{Love}}v^{10} \right]. \quad (17)$$

For the compact benchmark target, the point-particle baseline is kept at Newtonian order and the spin-induced finite-size pieces are retained:

$$e(v) = -\frac{\eta v^2}{2} \left[1 + \mathcal{E}_{\text{SS}}^{\text{SIM}}(v)v^4 + \mathcal{E}_{\text{SSS}}^{\text{SIM}}v^7 \right]. \quad (18)$$

The quadratic-in-spin piece is

$$\begin{aligned} \mathcal{E}_{\text{SS}}^{\text{SIM}}(v) = & \chi_a \chi_s \left(-\frac{\delta^2 \kappa_a}{2} - \delta \kappa_s - \frac{\kappa_a}{2} \right) \\ & + \chi_s^2 \left(-\frac{\delta^2 \kappa_s}{4} - \frac{\delta \kappa_a}{2} - \frac{\kappa_s}{4} \right) + \chi_a^2 \left(-\frac{\delta \kappa_a}{2} + \eta \kappa_s - \frac{\kappa_s}{2} \right) \\ & + v^2 \left[\chi_a^2 \left(\frac{25\delta\eta\kappa_a}{12} - \frac{35\delta\kappa_a}{12} - \frac{5\eta^2\kappa_s}{6} + \frac{95\eta\kappa_s}{12} - \frac{35\kappa_s}{12} \right) \right. \\ & + \chi_a \chi_s \left(\frac{25\delta\eta\kappa_s}{6} - \frac{35\delta\kappa_s}{6} - \frac{5\eta^2\kappa_a}{3} + \frac{95\eta\kappa_a}{6} - \frac{35\kappa_a}{6} \right) \\ & \left. + \chi_s^2 \left(\frac{25\delta\eta\kappa_a}{12} - \frac{35\delta\kappa_a}{12} - \frac{5\eta^2\kappa_s}{6} + \frac{95\eta\kappa_s}{12} - \frac{35\kappa_s}{12} \right) \right]. \end{aligned} \quad (19)$$

The cubic-in-spin piece is

$$\begin{aligned} \mathcal{E}_{\text{SSS}}^{\text{SIM}} = & \chi_a \chi_s^2 \left(-2\delta^2\eta\kappa_a - 5\delta^2\kappa_a + 6\delta\eta\kappa_s - 6\delta\kappa_s - \kappa_a \right) \\ & + \chi_s^3 \left(-\delta^3\kappa_a - \delta^2\eta\kappa_s - \frac{9\delta^2\kappa_s}{4} - \delta\kappa_a + \frac{\kappa_s}{4} \right) \\ & + \chi_a^2 \chi_s \left(-6\delta\kappa_a + 4\eta^2\kappa_s + 12\eta\kappa_s - 6\kappa_s \right) \\ & + \chi_a^3 \left(-2\delta\eta\kappa_s - 2\delta\kappa_s + 2\eta\kappa_a - 2\kappa_a \right). \end{aligned} \quad (20)$$

6 Flux Used in the Code

The paper writes the flux at infinity as

$$\mathcal{F}_\infty = \frac{32}{5}\eta^2 v^{10} \left[\mathcal{F}_{\text{NS}} + \mathcal{F}_{\text{SO}}v^3 + \mathcal{F}_{\text{SS}}v^4 + \mathcal{F}_{\text{SSS}}v^7 + \mathcal{F}_{\text{Love}}v^{10} \right]. \quad (21)$$

The benchmark implementation uses

$$\mathcal{F}_\infty = \frac{32}{5}\eta^2 v^{10} \left[1 + \mathcal{F}_{\text{SS}}^{\text{SIM}}(v)v^4 + \mathcal{F}_{\text{SSS}}^{\text{SIM}}v^7 + \frac{39}{8}\tilde{\Lambda}v^{10} \right]. \quad (22)$$

The $\tilde{\Lambda}$ term is an effective leading Love-number flux correction: with Newtonian binding energy, it reproduces the leading $-39\tilde{\Lambda}v^{10}/2$ tidal phase term quoted in the paper. A future production

waveform should replace this with the full $\mathcal{E}_{\text{Love}}$ and $\mathcal{F}_{\text{Love}}$ formulae from Flanagan–Iyer–Will, as referenced in Appendix A of the paper.

The quadratic spin-induced flux is

$$\begin{aligned}
\mathcal{F}_{\text{SS}}^{\text{SIM}}(v) = & \chi_a \chi_s (\delta^2 \kappa_a + 2\delta \kappa_s + \kappa_a) + \chi_a^2 (\delta \kappa_a - 2\eta \kappa_s + \kappa_s) \\
& + \chi_s^2 (\delta \kappa_a - 2\eta \kappa_s + \kappa_s) \\
& + v^2 \left[\chi_a^2 \left(-\frac{127\delta\eta\kappa_a}{16} + \frac{\delta\kappa_a}{14} + \frac{43\eta^2\kappa_s}{4} - \frac{905\eta\kappa_s}{112} + \frac{\kappa_s}{14} \right) \right. \\
& + \chi_a \chi_s \left(-\frac{127\delta\eta\kappa_s}{8} + \frac{\delta\kappa_s}{7} + \frac{43\eta^2\kappa_a}{2} - \frac{905\eta\kappa_a}{56} + \frac{\kappa_a}{7} \right) \\
& + \chi_s^2 \left(-\frac{127\delta\eta\kappa_a}{16} + \frac{\delta\kappa_a}{14} + \frac{43\eta^2\kappa_s}{4} - \frac{905\eta\kappa_s}{112} + \frac{\kappa_s}{14} \right) \left. \right] \\
& + v^3 \left[\chi_a^2 (4\pi\delta\kappa_a - 8\pi\eta\kappa_s + 4\pi\kappa_s) \right. \\
& + \chi_a \chi_s (8\pi\delta\kappa_s - 16\pi\eta\kappa_a + 8\pi\kappa_a) \\
& + \chi_s^2 (4\pi\delta\kappa_a - 8\pi\eta\kappa_s + 4\pi\kappa_s) \left. \right]. \tag{23}
\end{aligned}$$

The cubic spin-induced flux is

$$\begin{aligned}
\mathcal{F}_{\text{SSS}}^{\text{SIM}} = & \chi_a \chi_s^2 \left(\frac{13}{3} \delta^2 \eta \kappa_a + \frac{27}{16} \delta^2 \kappa_a + \frac{4}{3} \delta \eta \kappa_s + \frac{15}{8} \delta \kappa_s + \frac{3}{16} \kappa_a \right) \\
& + \chi_a^2 \chi_s \left(\frac{95}{12} \delta \eta \kappa_a + \frac{15}{8} \delta \kappa_a - \frac{26}{3} \eta^2 \kappa_s + \frac{25}{6} \eta \kappa_s + \frac{15}{8} \kappa_s \right) \\
& + \chi_s^3 \left(-\frac{7}{4} \delta \eta \kappa_a + \frac{5}{8} \delta \kappa_a - \frac{26}{3} \eta^2 \kappa_s - 3\eta \kappa_s + \frac{5}{8} \kappa_s \right) \\
& + \chi_a^3 \left(\frac{29}{6} \delta \eta \kappa_s + \frac{5}{8} \delta \kappa_s + \frac{43}{12} \eta \kappa_a + \frac{5}{8} \kappa_a \right). \tag{24}
\end{aligned}$$

The absorbed flux is

$$\begin{aligned}
\dot{M}(v) = & \frac{1}{2} \left(9\bar{\mathcal{H}}_1^E \eta^2 \chi_a + 9\mathcal{H}_1^E \eta^2 \chi_s \right) v^{15} \\
& + \frac{1}{2} \left[\left(9\mathcal{H}_1^B \eta^2 + \frac{45}{2} \mathcal{H}_1^E \eta^2 + \frac{9}{2} \bar{\mathcal{H}}_1^E \delta \eta^2 - 27\mathcal{H}_1^E \eta^3 \right) \chi_s \right. \\
& + \left. \left(9\bar{\mathcal{H}}_1^B \eta^2 + \frac{45}{2} \bar{\mathcal{H}}_1^E \eta^2 + \frac{9}{2} \mathcal{H}_1^E \delta \eta^2 - 27\bar{\mathcal{H}}_1^E \eta^3 \right) \chi_a \right] v^{17} \\
& + 18\mathcal{H}_0 \eta^2 v^{18}. \tag{25}
\end{aligned}$$

Appendix A prints $9\mathcal{H}_0 \eta^2 v^{18}$. The code default doubles that term to implement the BBH electric–magnetic duality convention used by the benchmark. Setting `h0_dual=False` restores the electric-only coefficient.

7 Frequency Band and Taper

The waveform is inspiral-only. The cutoff is

$$f_{\text{ISCO}} = \frac{1}{6^{3/2}\pi M_{\text{sec}}}, \quad f_{\text{cut}} = q_{\text{max}} f_{\text{ISCO}}, \quad (26)$$

where $q_{\text{max}} = \text{fmax_over_fisco}$, with default $q_{\text{max}} = 1$. The waveform is exactly zero for $f < f_{\text{low}}$ or $f \geq f_{\text{cut}}$. Inside the band the taper is

$$W(f) = \frac{1}{1 + \exp[(f - f_{\text{cut}})/\sigma]}, \quad \sigma = \text{sigma_taper_over_fisco} f_{\text{ISCO}}. \quad (27)$$

The default is $\text{sigma_taper_over_fisco} = 0.01$.

8 Code API

The benchmark-facing function is

```
h_of_f(
    f, Mc, eta, dL,
    chi1=0.0, chi2=0.0,
    tc=0.0, phic=0.0,
    kappa_s=1.0, kappa_a=0.0,
    H0=0.0, Lambda_tilde=0.0,
    H1E=0.0, H1B=0.0,
    H1E_bar=0.0, H1B_bar=0.0,
    f_low=20.0,
    fmax_over_fisco=1.0,
    sigma_taper_over_fisco=0.01,
    phase_only=False,
    h0_dual=True,
)
```

It returns a complex NumPy array with the same shape as the input frequency array. If `phase_only=True`, the code returns $W(f) \exp[-i\psi(f)]$ inside the band and zero outside it.

9 Current Caveats

- The point-particle baseline in the balance-law dynamics is Newtonian. Higher-PN point-particle terms can be added later by replacing the $\mathcal{E}_{\text{NS}}$, $\mathcal{E}_{\text{SO}}$, $\mathcal{F}_{\text{NS}}$, and $\mathcal{F}_{\text{SO}}$ blocks.
- The Love-number sector is represented by the leading effective flux term in Eq. (22). This keeps the benchmark compact, but it is not the full Flanagan–Iyer–Will tidal energy and flux model.
- The cutoff is f_{ISCO} -based, not the paper’s $f_{22}^{\text{tape}} = 0.35 f_{22}^{\text{peak}}$ prescription.
- The implementation is intended for benchmark reproducibility and forecasting experiments, not final astrophysical inference.